Modeling the Unreal

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Outline

- Realism
- Information & Aboutness
- Three Problem Cases
- About the Unreal

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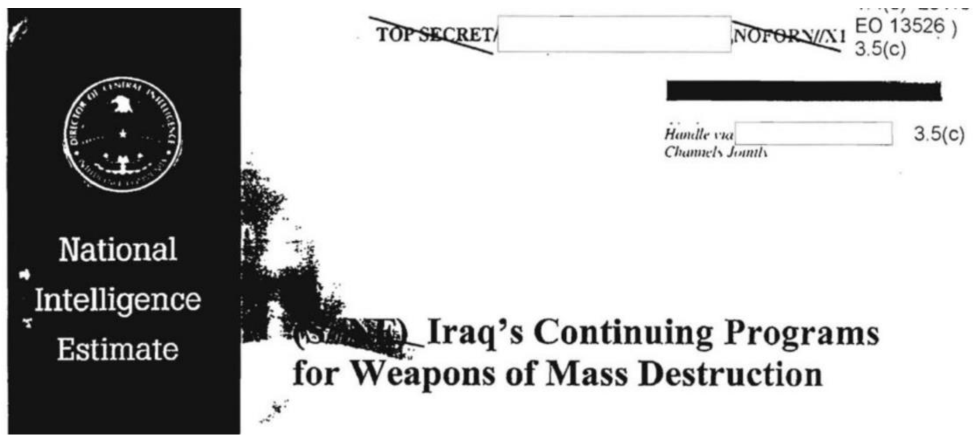
Directions of Fit for Ontology Engineering

- Ontology engineering strategies can be distinguished by direction of fit
  - **World-to-Word:** Make world fit the words
  - **Word-to-World:** Make words fit the world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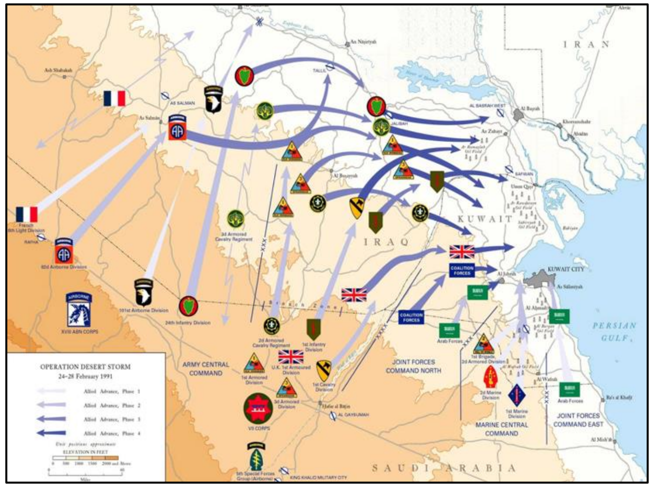
National Intelligence Estimate

~~(S//NF)~~ Iraq's Continuing Programs for Weapons of Mass Destruction



Direction of Fit

World-to-Word: Make world fit the words

Word-to-World: Make words fit the world

Suppose you're a spy watching someone shop in a grocery store. They have a shopping list and are checking items off as they put them in their basket.

Suppose his list directs him to buy butter

Suppose he instead buys margarine

Direction of Fit

World-to-Word: Make world fit the words

Word-to-World: Make words fit the world

Suppose you are aiming to align his behavior with the guidance on the shopping list

One way to do that would be to erase "butter" on the list and replace it with "margarine"

Making the words on the list align with his behavior in the world

Direction of Fit

World-to-Word: Make world fit the words

Word-to-World: Make words fit the world

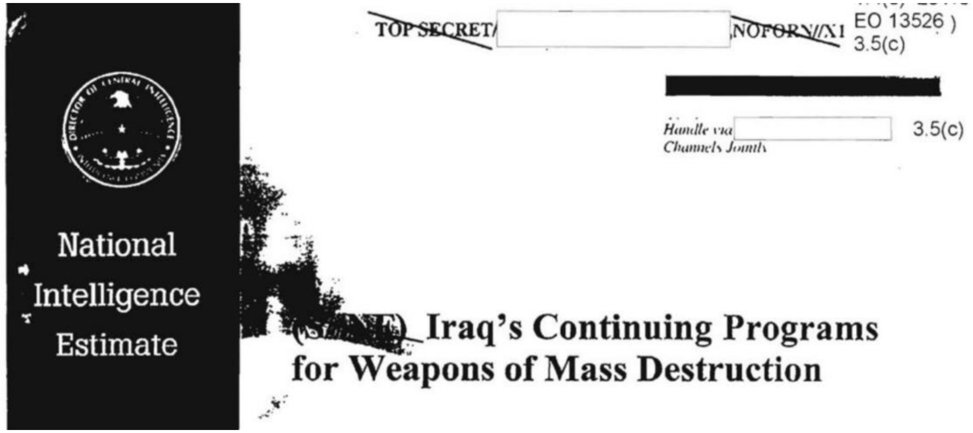
Suppose you are instead aiming to record his behavior as accurately as possible

One way to do that would be to record the divergence of his behavior from the list

Making his behavior in the world clealy deviate from the words on the list

Directions of Fit for Ontology Engineering

- Ontology engineering strategies can be distinguished by direction of fit
  - **World-to-Word:** Make world fit the words, e.g. **conceptualism**
  - **Word-to-World:** Make words fit the world, e.g. **realism**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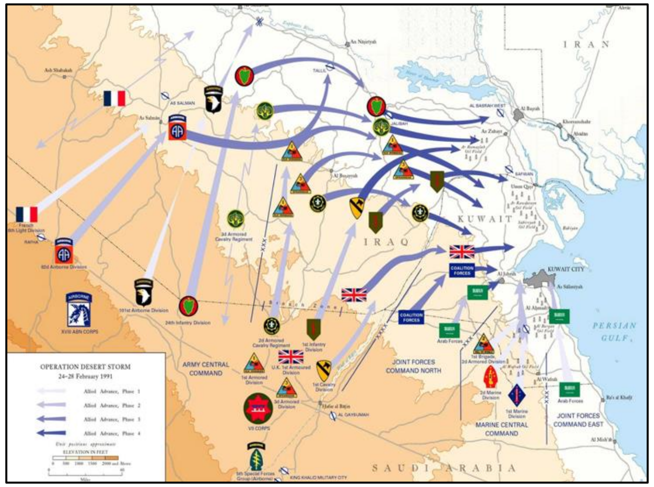

Conceptualism

- "An ontology is a specification of a conceptualization" –*Gruber, 1992*
- Interpreted as claiming that ontologies represent **concepts**
- "Taking a more pragmatic view, one can say that ontology is a tool and product of engineering and thereby defined by its use." –*Gruber, 2009*
- Interpreted as claiming ontologies represent **what they need to represent**

Realism

- Ontologies are a specification of **the world**

If we attempt to go from the words we use to the world, it is unlikely that we will end up in the same place

If we attempt to go from the world to the words we use, it is more likely we will remain coordinated

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Pattern Recognition

- We're disposed to recognize patterns with our perceptual faculties…

Pattern Recognition

- We're disposed to recognize patterns with our perceptual faculties…

- …and with our cognitive faculties…

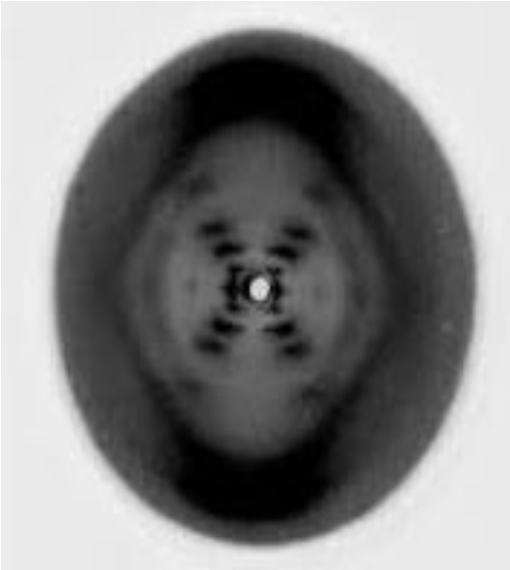


Photo 51 crystallography image by Rosalind

Double-helix hypothesized by Watson and Crick

Real Patterns

- Some patterns are necessarily **about** something; some patterns are not
- "Snow is white" expresses content that is **about** snow
- "cm" or "." are not necessarily **about** anything; they are nevertheless patterns
- Most patterns of interest to ontologists are **about** something

Aboutness

- **Information** is a pattern that is **about** something

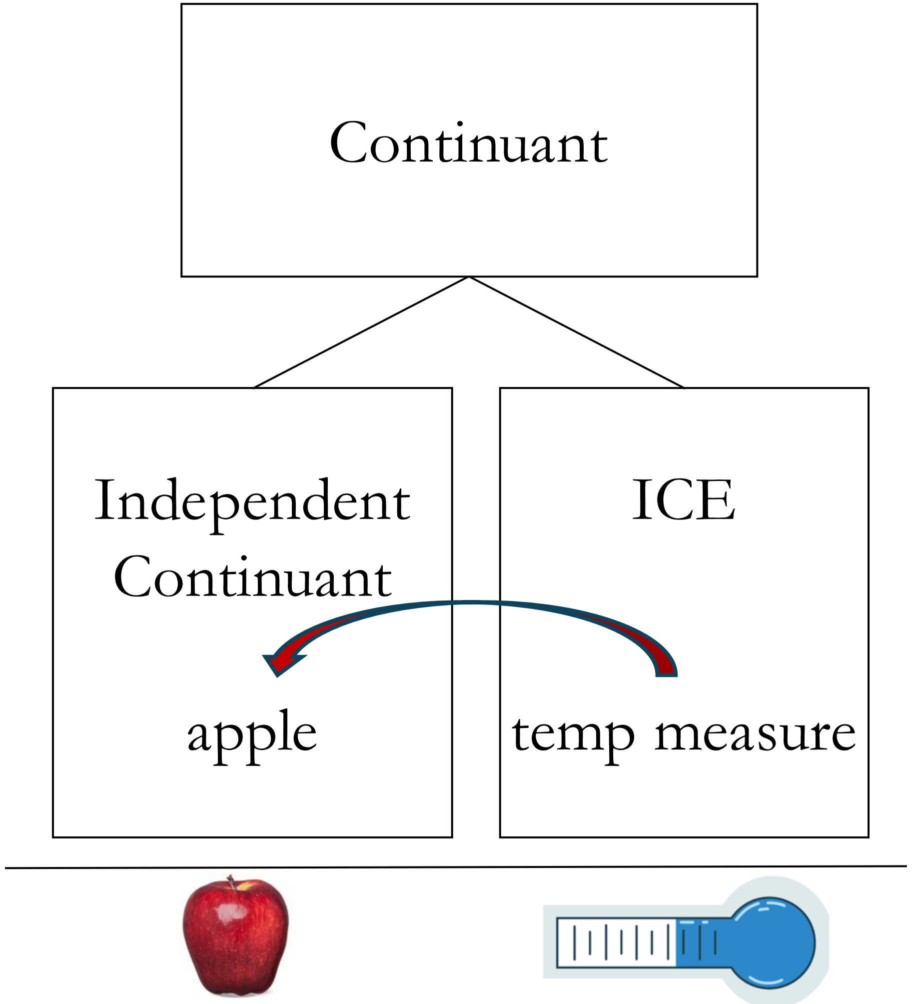
- In BFO extensions - such as the Information Artifact Ontology and the Information Entity Ontology - information is represented by the class **Information Content Entity**

- Where the "is about" relation is understood to be primitive:

definition [language: en]

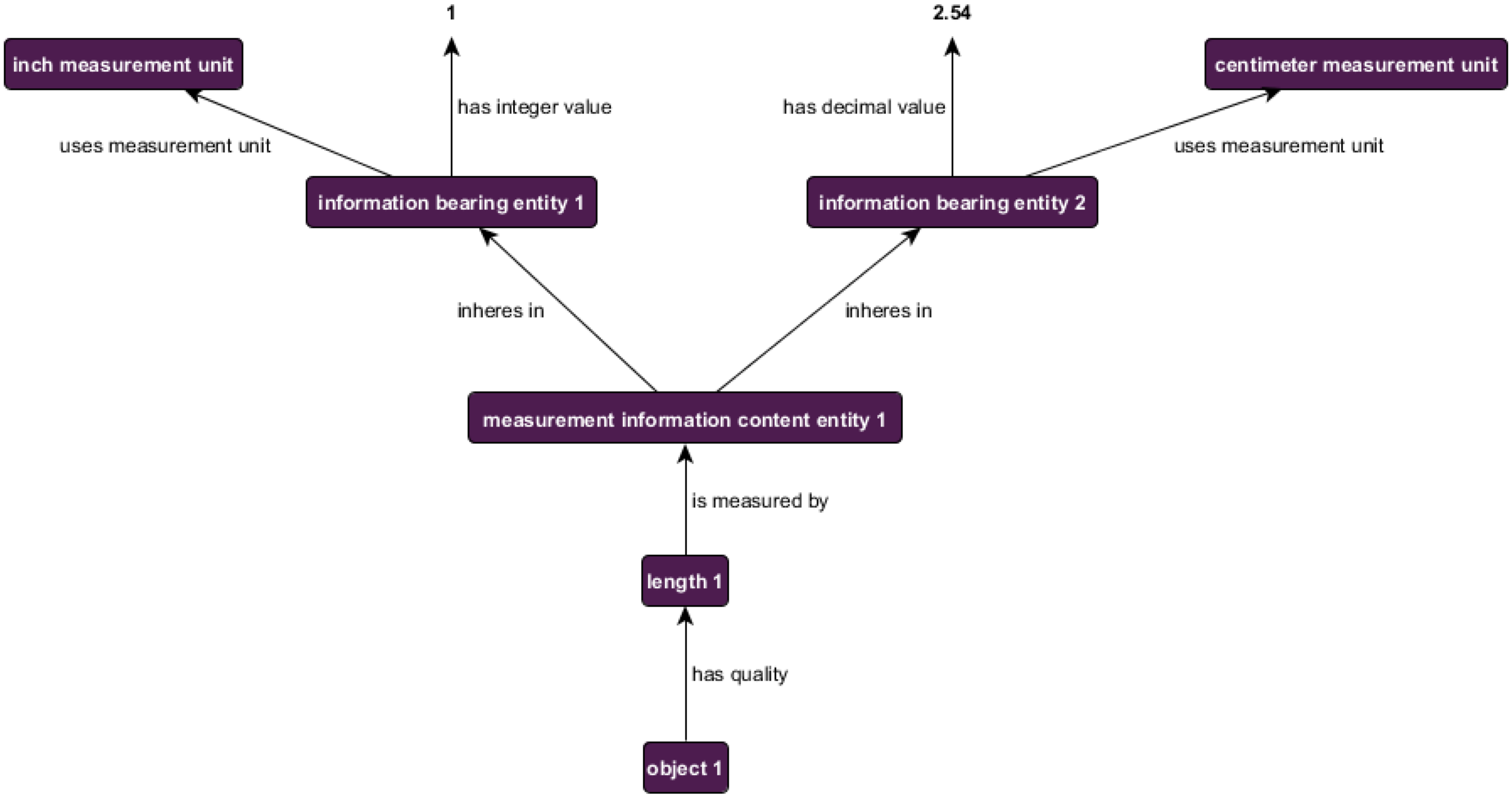
A primitive relationship between an Information Content Entity and some Entity.

Information Content Entity

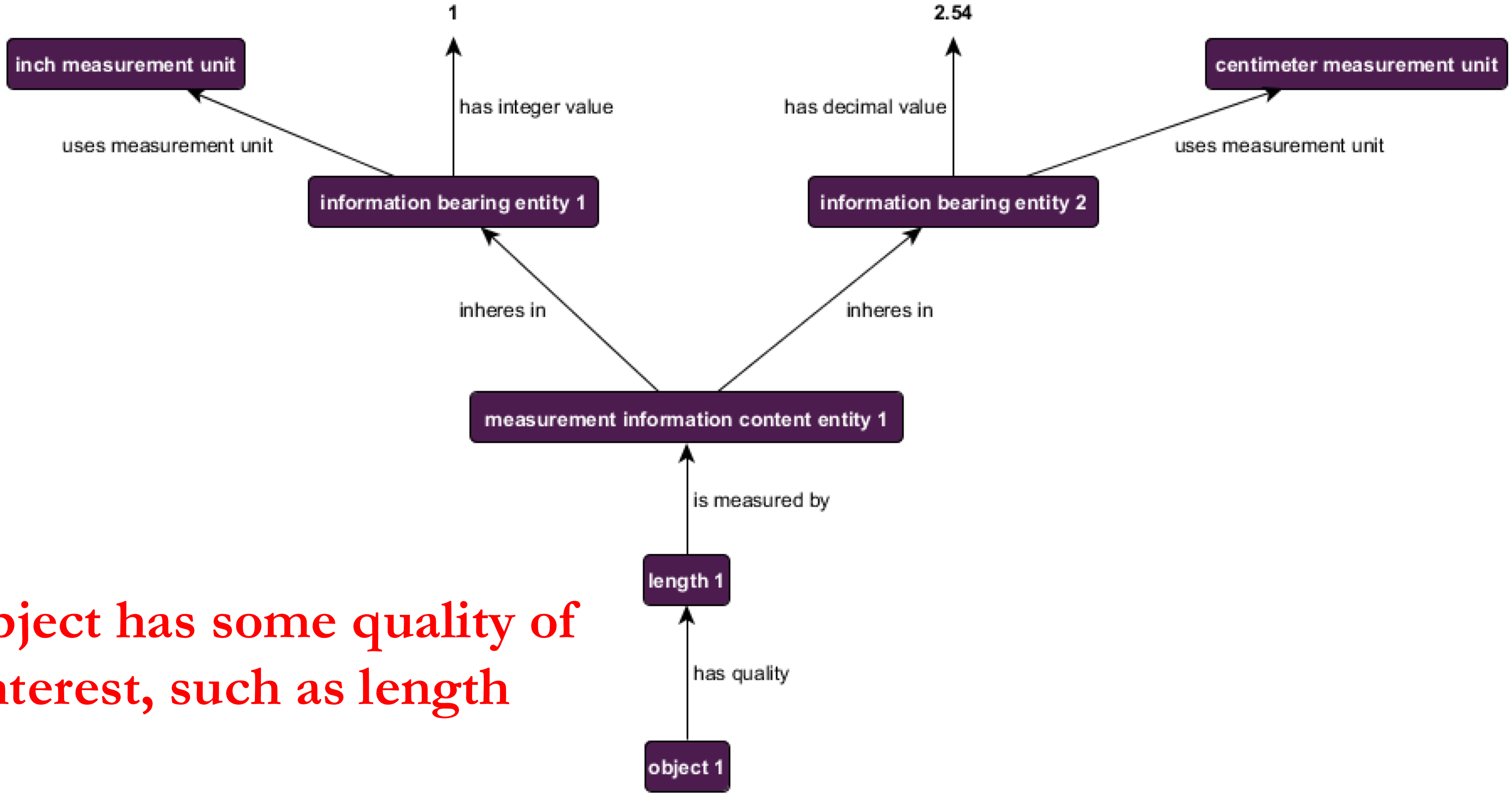


For example, many measures are about independent continuants

CCO Measurement



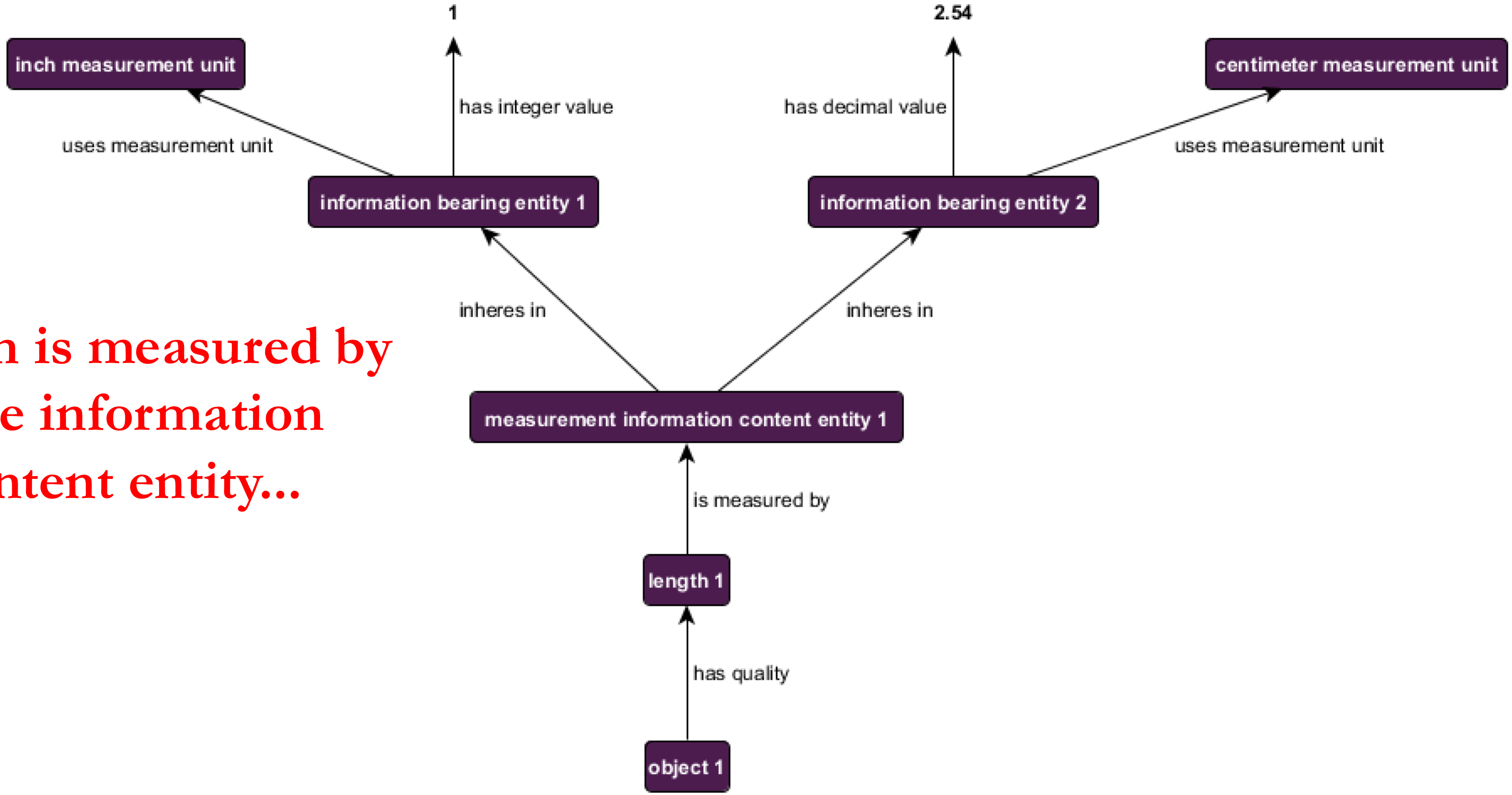
CCO Measurement



An object has some quality of interest, such as length

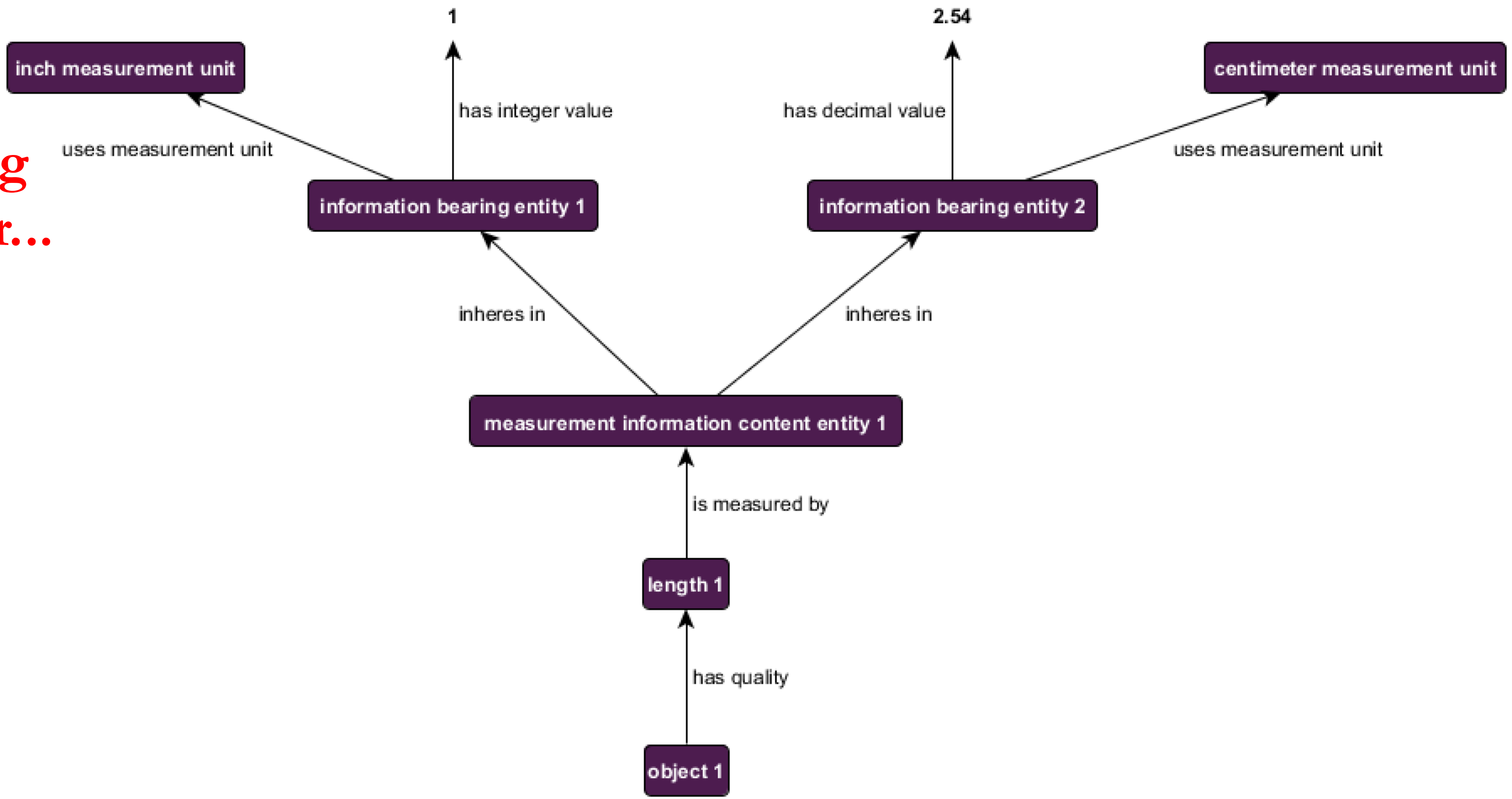
CCO Measurement

…which is measured by some information content entity…



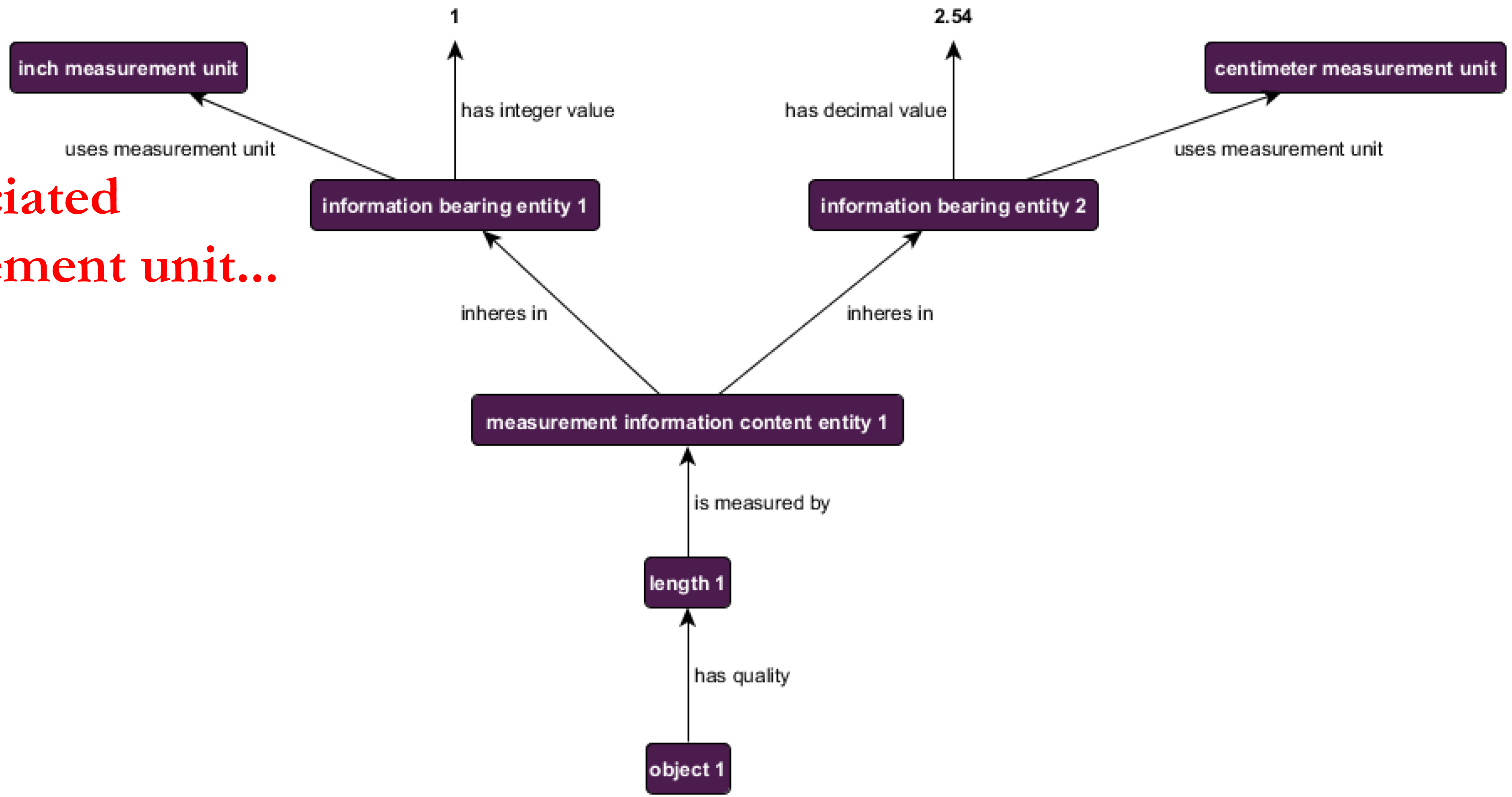
CCO Measurement

...inhering in a bearer...



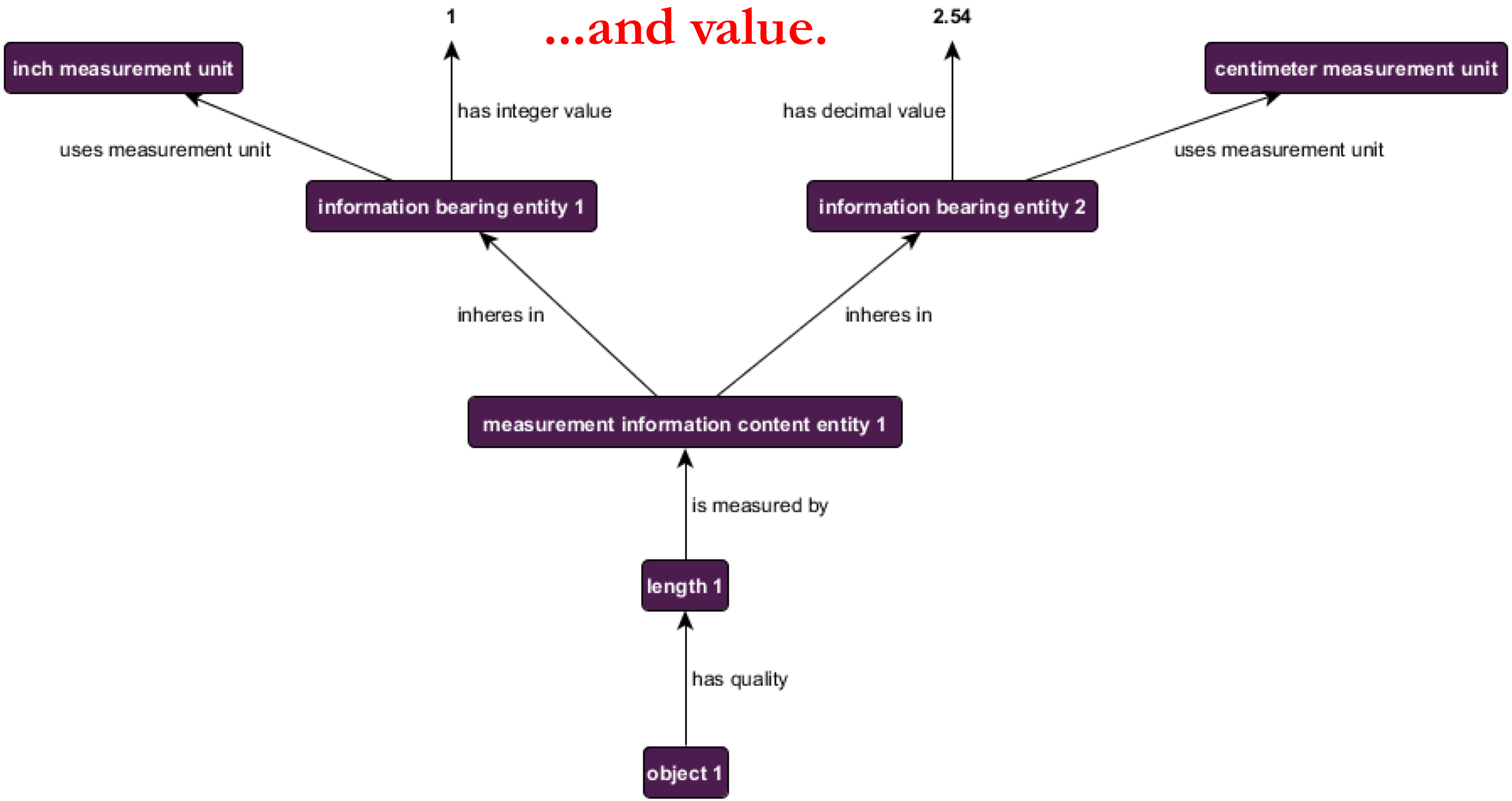
CCO Measurement

...with
an associated
measurement unit...

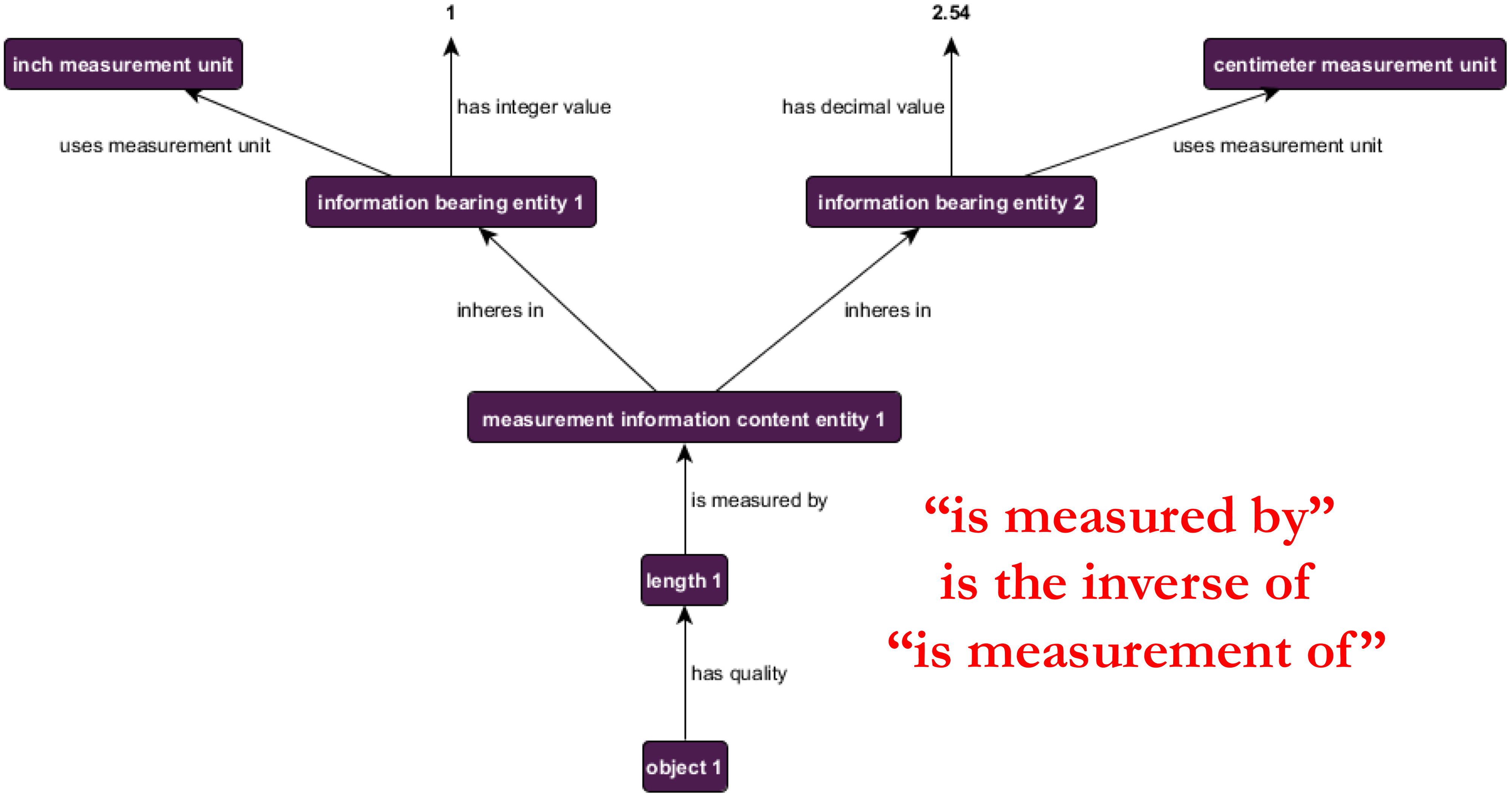


CCO Measurement

...and value.

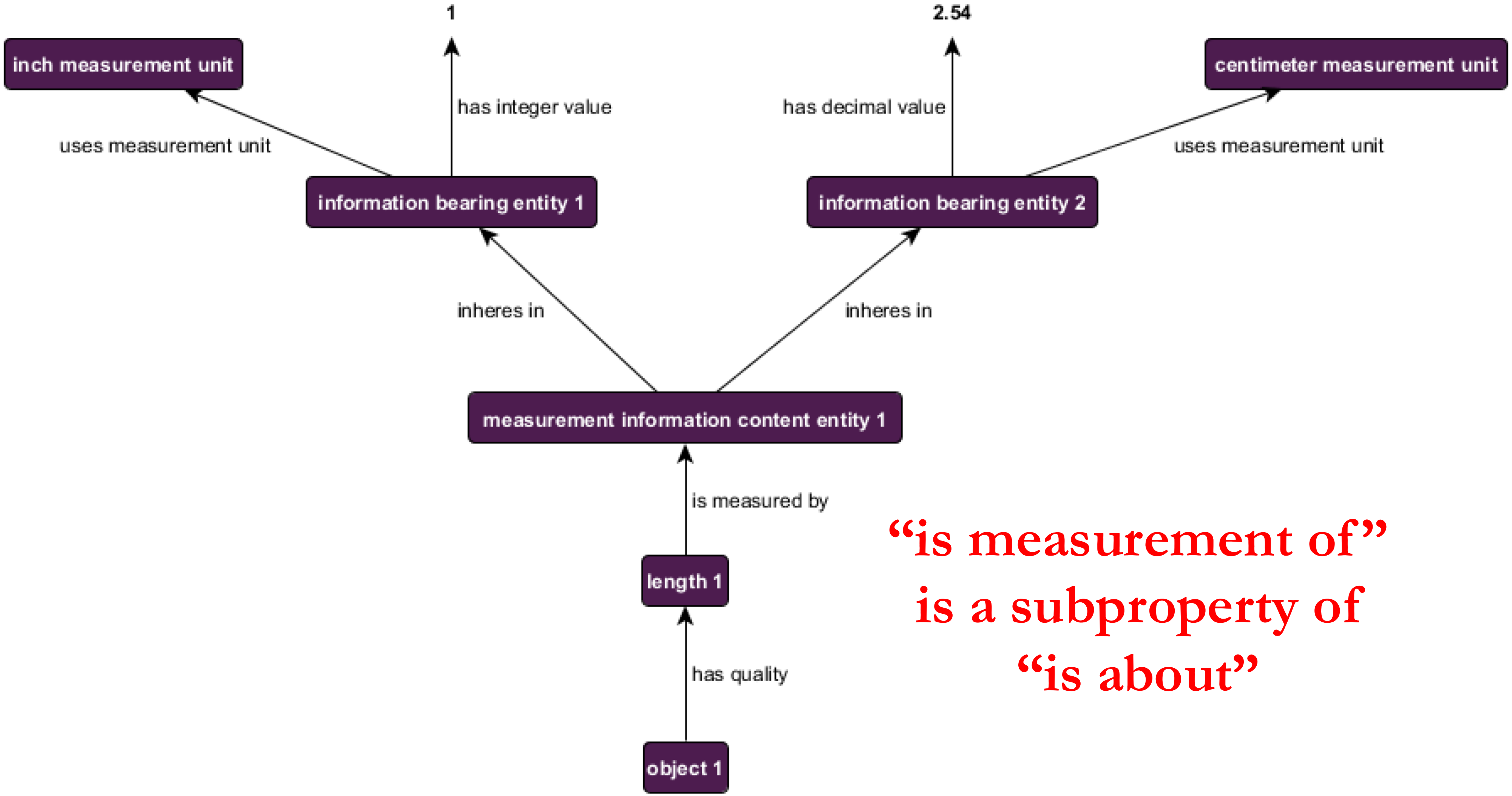


CCO Measurement



"is measured by"
is the inverse of
"is measurement of"

CCO Measurement



“is measurement of” is a subproperty of “is about”

Genericaly Dependent Continuant

Continuant

GD
Continuant

orchestra plan

GD
Continuant

symphony

Occurrent

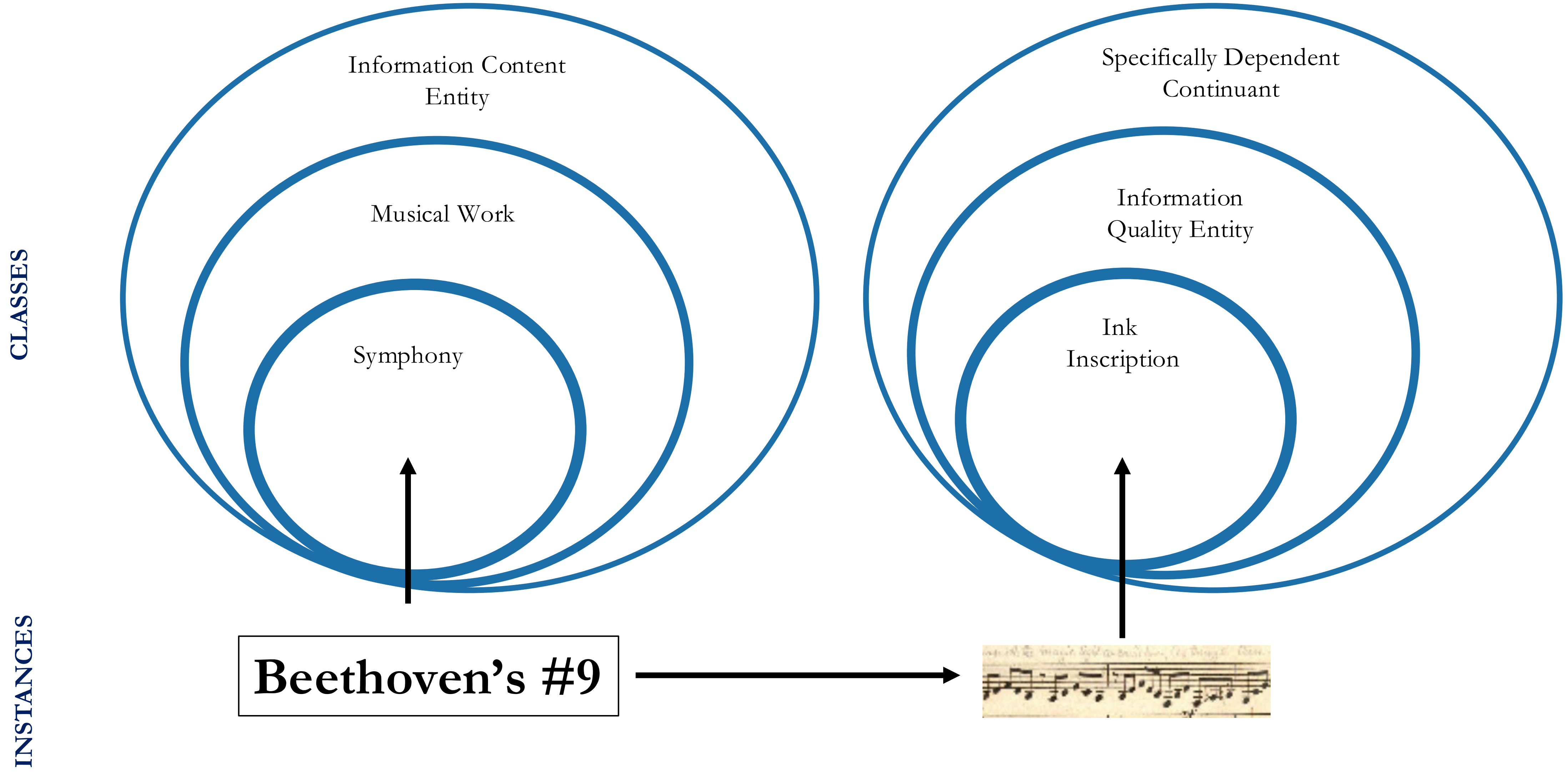
process, event

GDCs may be about other GDCs

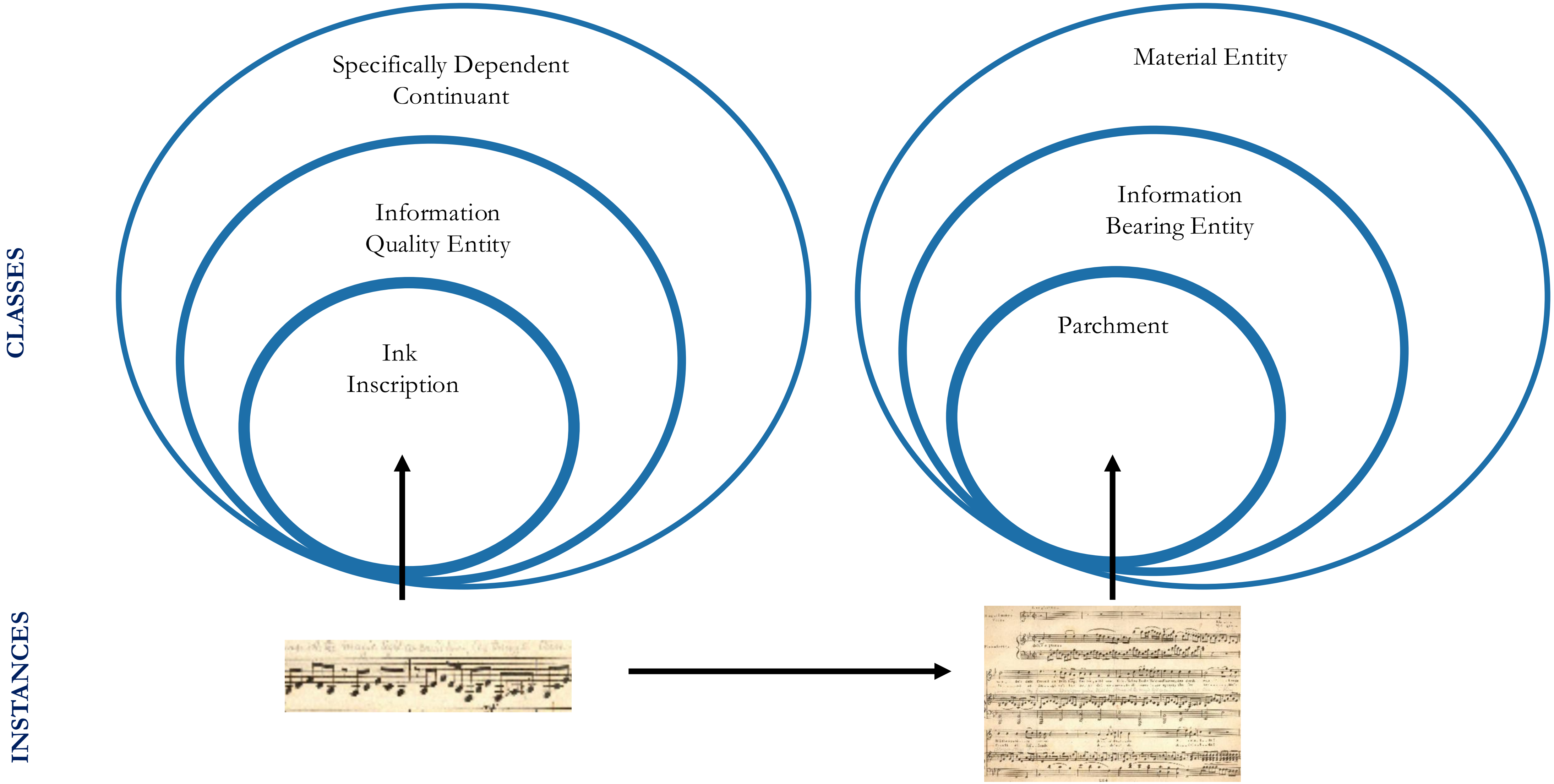
Beethoven's #9

• • • • • •

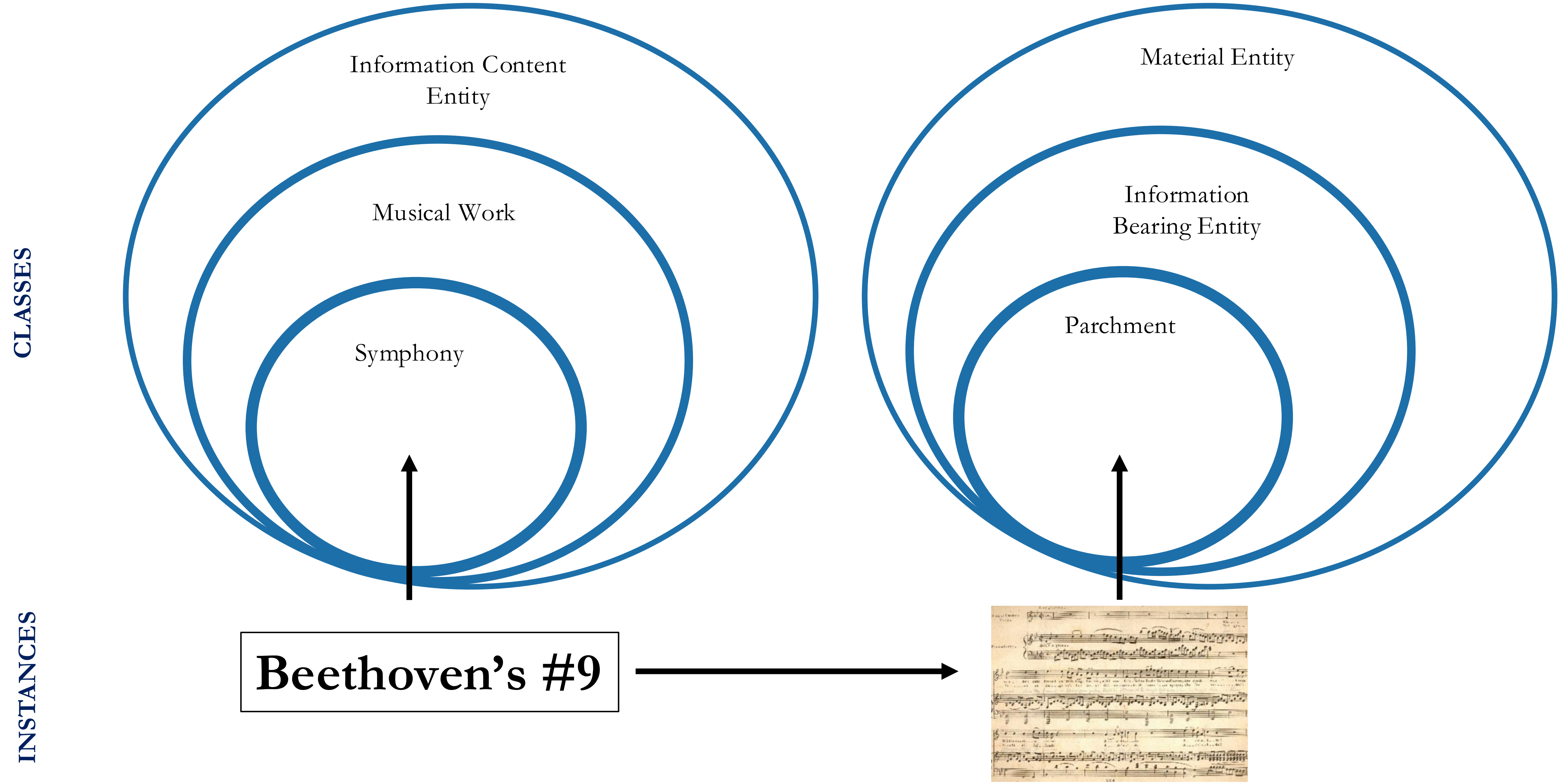
#9 Concretized in Ink Inscription



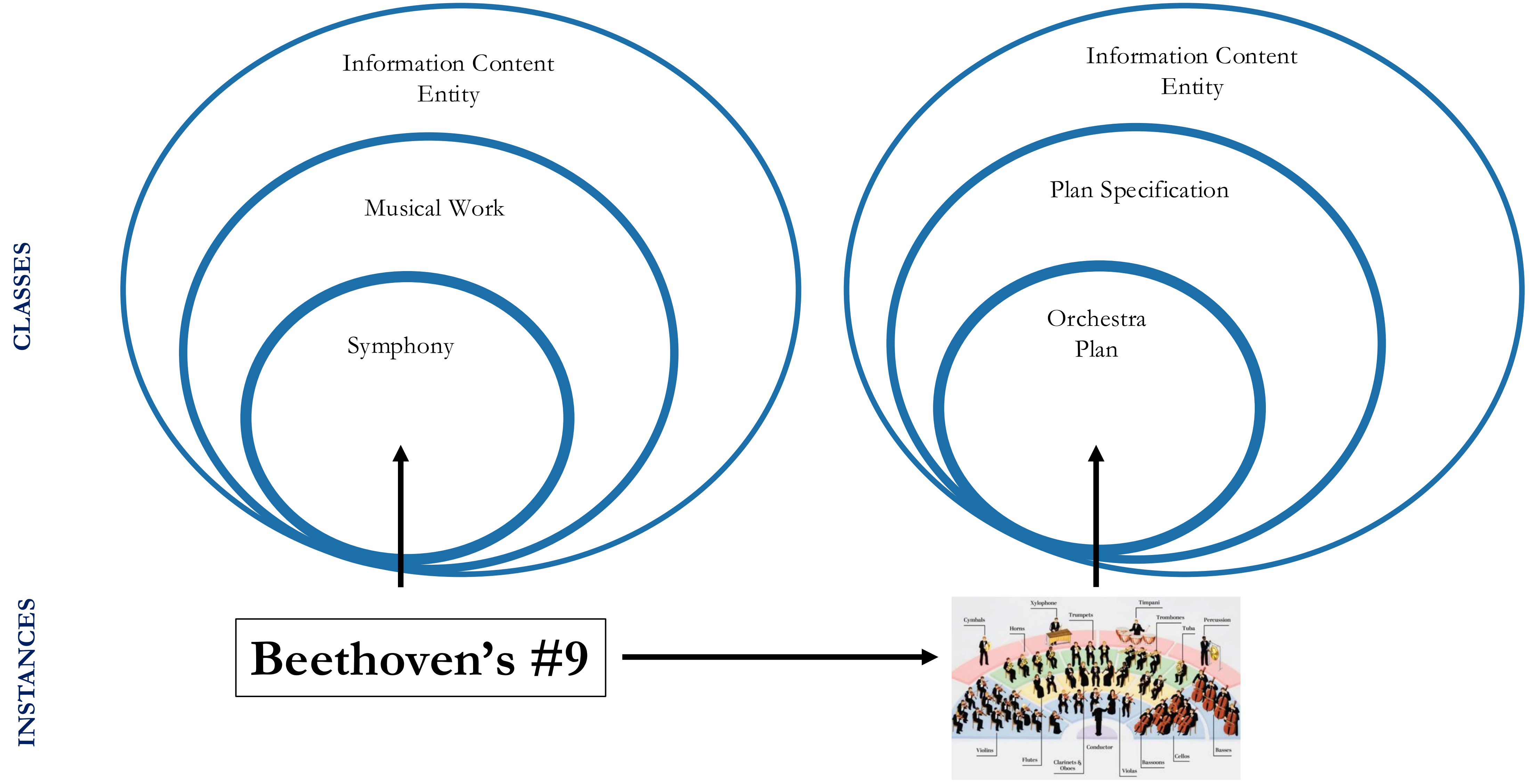
Ink Inscription inheres in Parchment



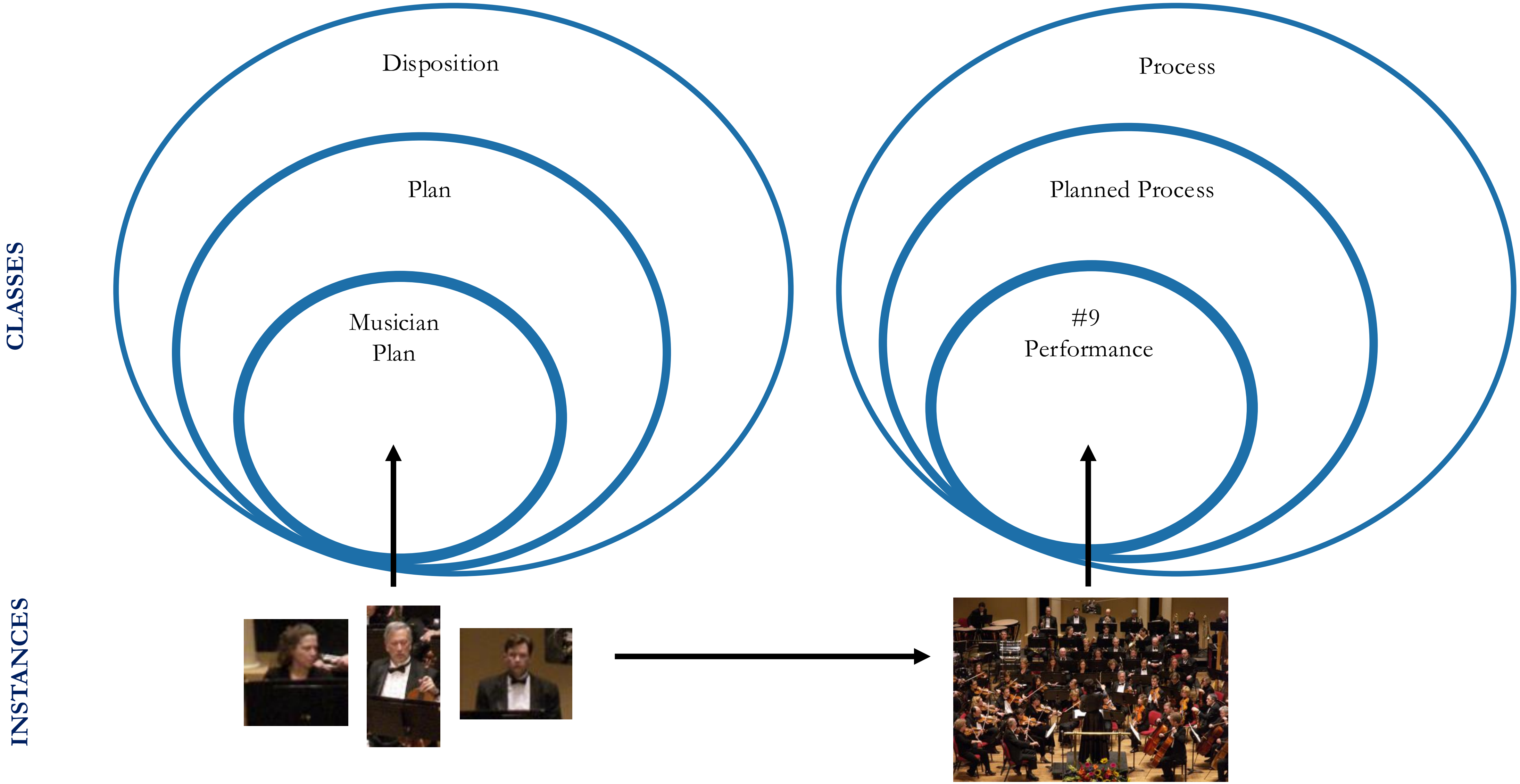
#9 Generically Dependent On Parchment



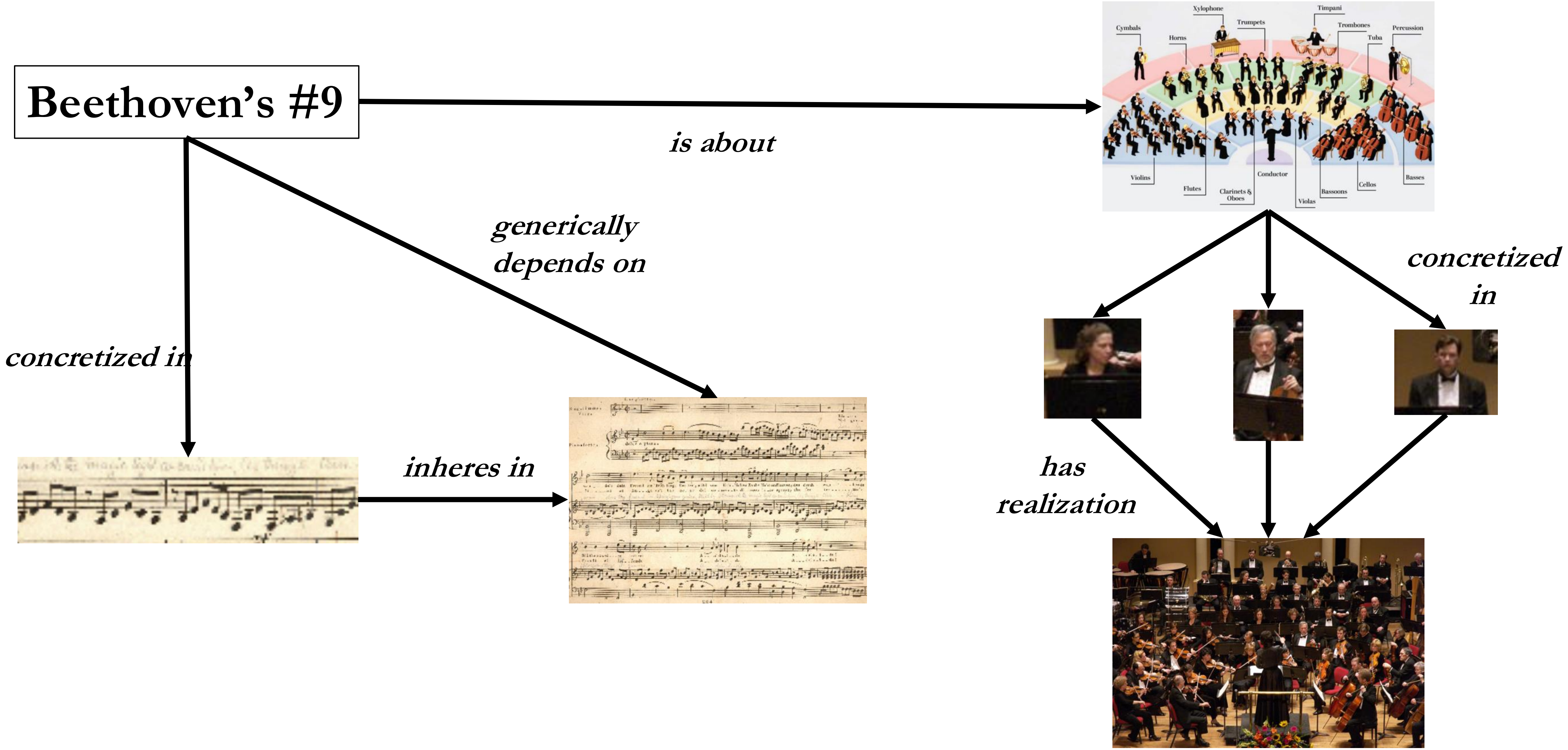
#9 is about Orchestra Plan



Musician Plan realized in #9 Performance



Beethoven’s #9th Symphony



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Three Cases

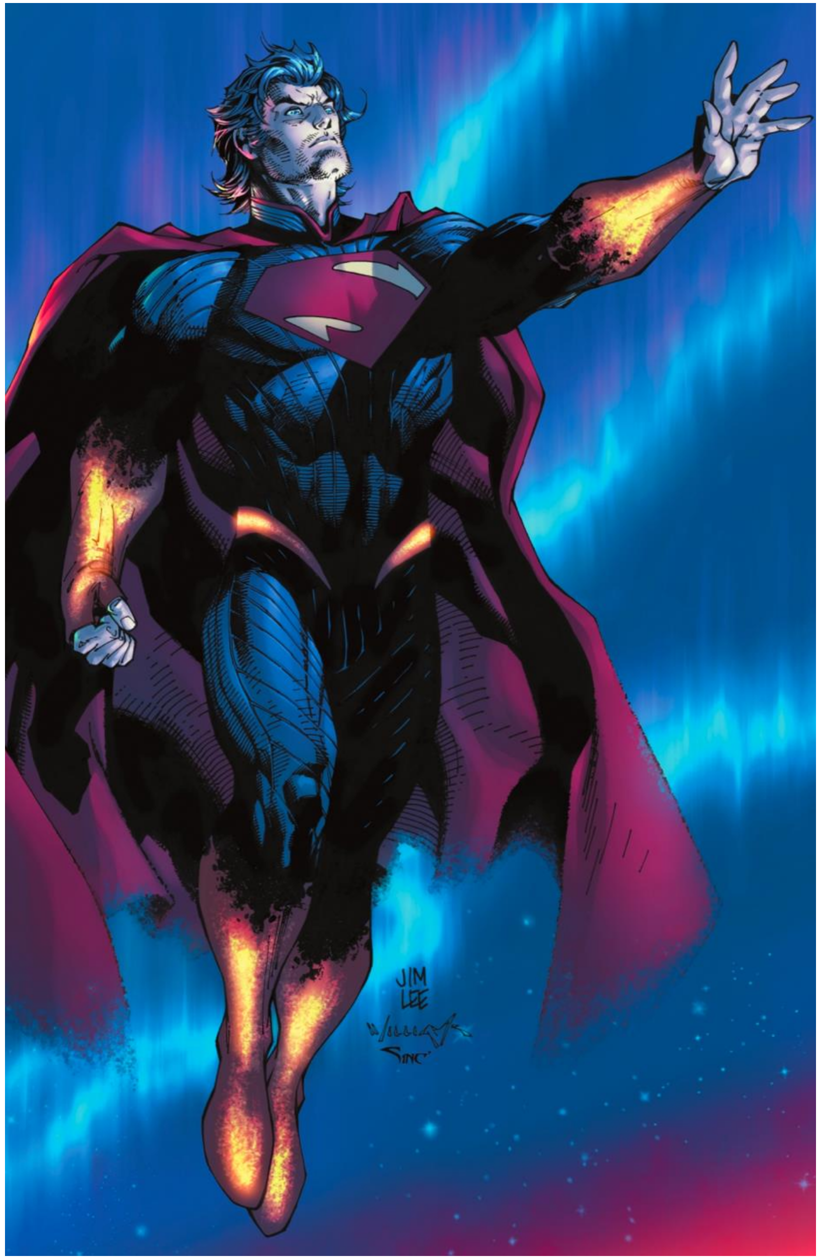
- In some cases it is tricky to determine what information is about
- **Fiction** – Sam claims that “Superman is stong”
- **Blueprints** – Samantha creates a blueprint for an “Airbus 320N”
- **Simulation** – Cybersecurity experts create red-team exercise “6126”

Similarities & Differences

- Each suggests there is something the given expression is about, e.g. "Superman is strong"

- One way of understanding this is in terms of concepts but this does not seem to do justice to the sense of Superman at play

- "Superman" refers not simply to your concept of Superman or mine, but something they have in common



Realism

- Realism means that information **must ultimately be about** real entities; however:

    - There is **no instance** which “Superman” is about
    - There is **no instance** which “Airbus 320N spec” is about
    - There is **no instance** which “Cyber Simulation 626” is about

- Yet these are clearly important cases to represent

Realism

- For an **instance to satisfy realism** it must correspond to some individual in the world; for a **class to satisfy realism** there must be one or more instances falling under it
- A **real class** is one that has instances, e.g. River, Mammal, etc.
- Since there are no instances beyond concepts in our three problem cases, we must rely on real classes

OWL

- This is a bigger challenge than realism, however, since OWL requires that relations hold **only** between instances

- Which is to say that for any given information that **is about** some y, that y must be an instance

- Bringing us back to the suggestion that claims such as "Superman is strong" in OWL must denote instances, despite there not being any such

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Fly Out of the Bottle

- This is, however, too quick

- OWL provides resources by which we can relate relevant information to logical combinations of **types of classes** without commitment to there being a specific instance at the intersection of the combination

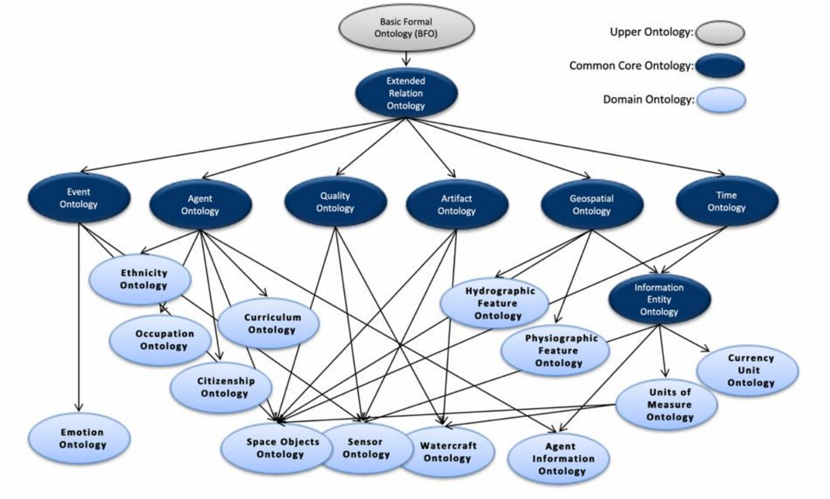
- Moreover, these logical combinations of classes can be decomposable ultimately into **real classes**, satisfying realism

Approach

- Model statements as **Information Content Entities** about logical combinations (intersection, union, negation) of **real classes**

- Adding to the story, leverage Common Core Ontologies to provide sub-properties of **is about** to distinguish these cases:

  - Describes (fiction)
  - Prescribes (blueprints)
  - Represents (simulations)



CCO Properties

- x *describes* y iff x is an **information content entity**, and y is an **entity**, such that x is about the characteristics by which y can be recognized or visualized

- x *prescribes* y iff x is an **information content entity** and y is an **entity**, such that x serves as a rule or guide for y if y is an **occurrent**, or x serves as a model for y if y is a **continuant**

- x *represents* y iff x is an **information content entity**, y is an **entity**, and z is a carrier of x, such that x is about y in virtue of there existing an isomorphism between characteristics of z and y

Recipe

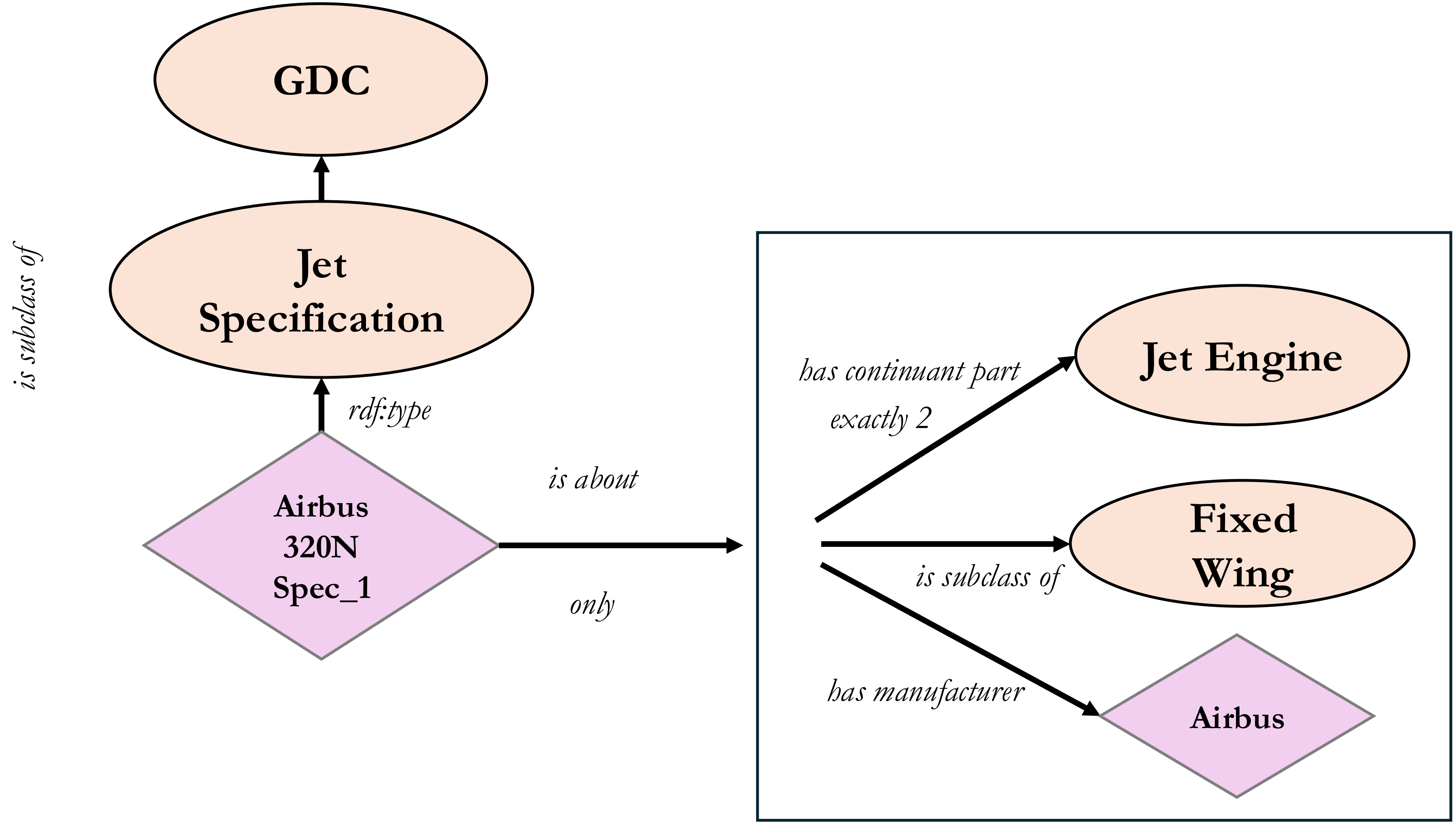
- Introduce an **information content entity** (ICE); identify classes and object properties reflecting intended meaning of the ICE such as **super strength**, *continuant part of*, and so on

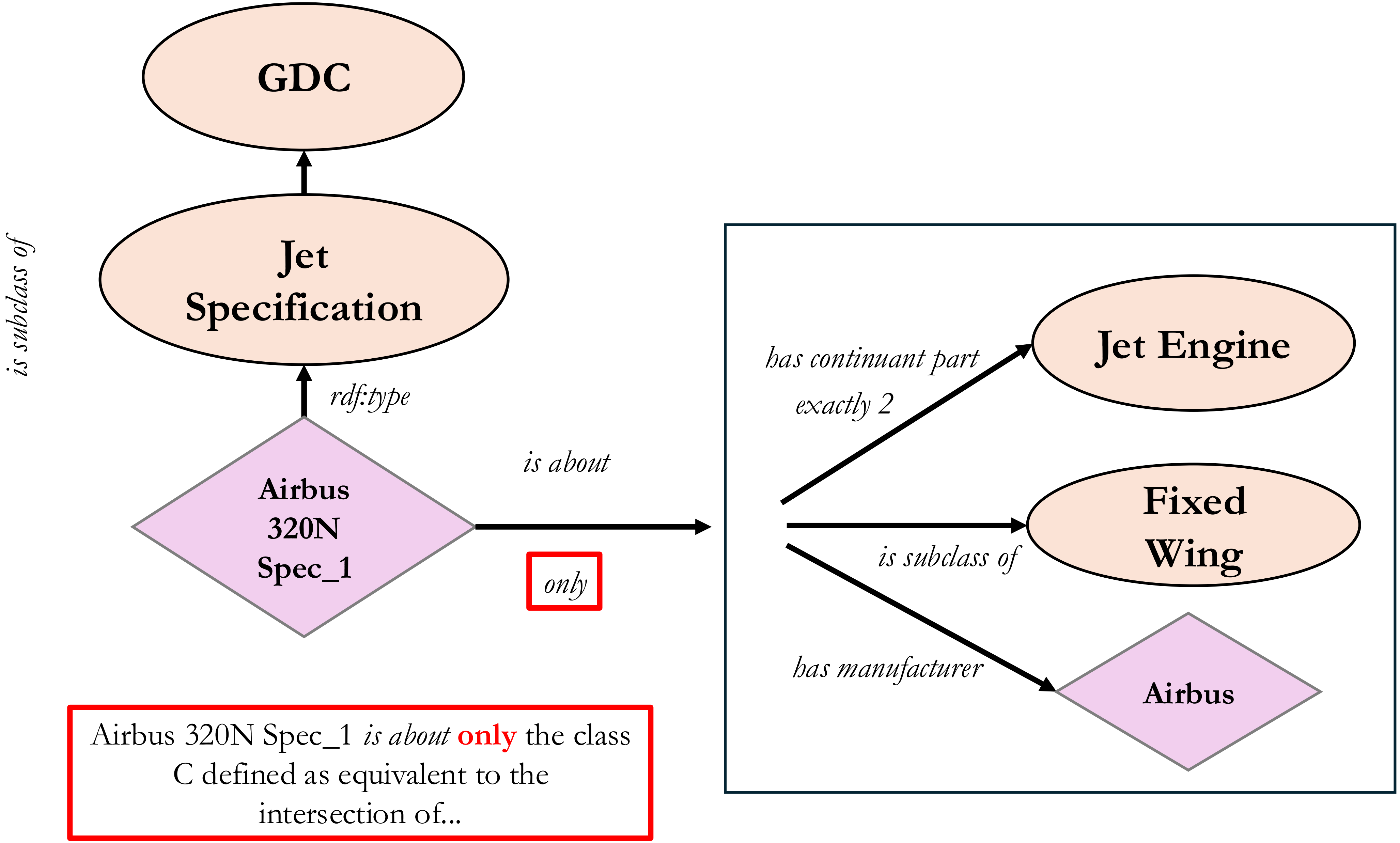
- Assert ICE *is about* **only** a class C equivalent to the logical combination of classes and restrictions on object properties where:
  - Fictions *describe*
  - Blueprints *prescribe*
  - Simulations *represent*

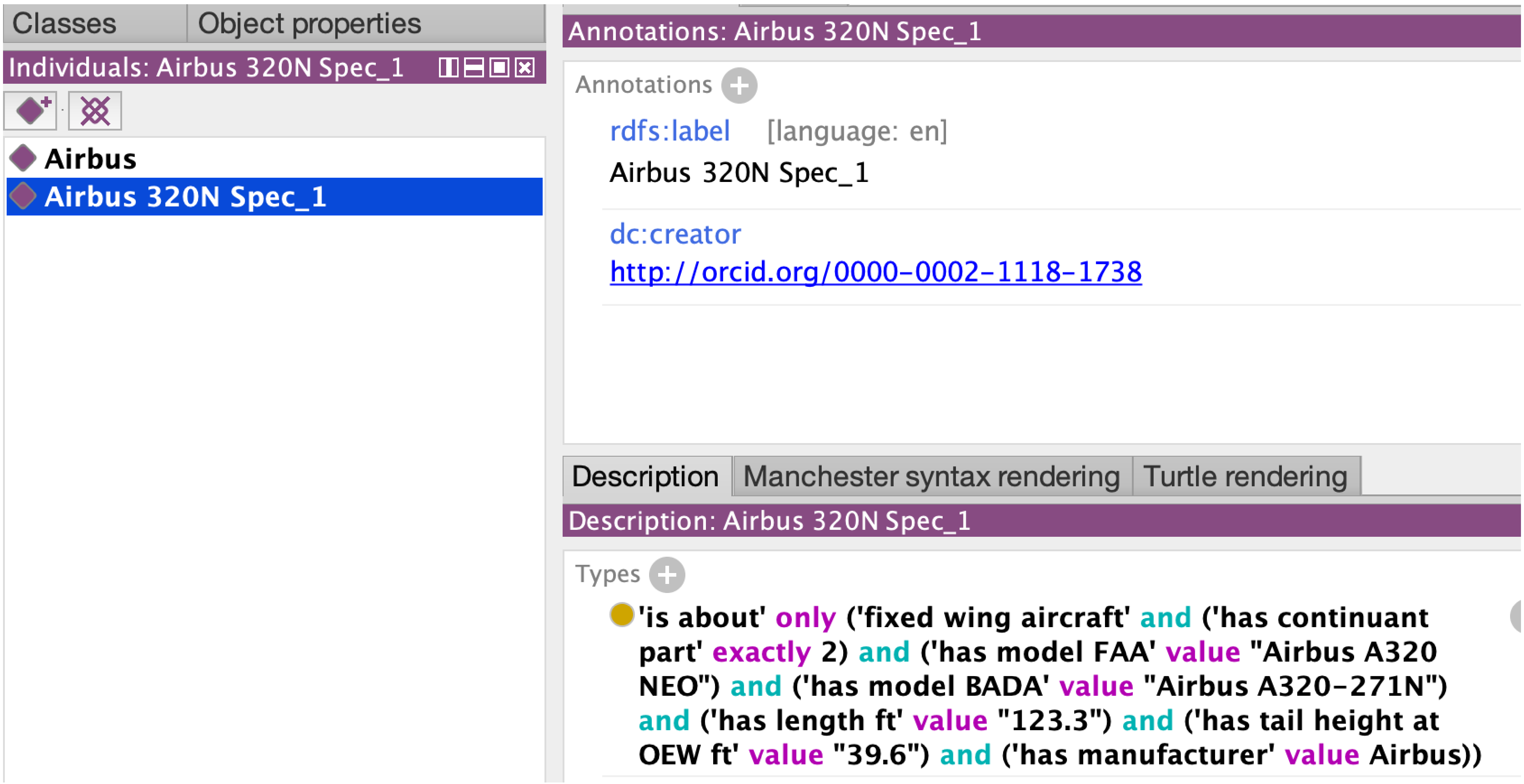
- For class or relata of any object property constituting C with no instance, return to second step; otherwise, stop

Airbus 320N Example

Samantha drafts a jet specification blueprint for an Airbus 320N attaching various model and manufacturer information, as well as requiring it be a fixed wing, have exactly two engines, Airbus as a manufacturer, and so on


GDC
is subclass of
Jet Specification
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part exactly 2
Jet Engine
Fixed Wing
is subclass of
has manufacturer
Airbus


GDC
is subclass of
Jet Specification
rdf:type
Airbus 320N Spec_1
is about
only
has continuant part
exactly 2
Jet Engine
Fixed Wing
is subclass of
has manufacturer
Airbus
Airbus 320N Spec_1 *is about* **only** the class C defined as equivalent to the intersection of...


Classes
Object properties
Individuals: Airbus 320N Spec_1
Airbus
Airbus 320N Spec_1
Annotations: Airbus 320N Spec_1
Annotations
rdfs:label [language: en]
Airbus 320N Spec_1
dc:creator
http://orcid.org/0000-0002-1118-1738
Description
Manchester syntax rendering
Turtle rendering
Description: Airbus 320N Spec_1
Types
'is about' only ('fixed wing aircraft' and ('has continuant part' exactly 2) and ('has model FAA' value "Airbus A320 NEO") and ('has model BADA' value "Airbus A320-271N") and ('has length ft' value "123.3") and ('has tail height at OEW ft' value "39.6") and ('has manufacturer' value Airbus))

Superman Example

Superman instance of **Fictional Description** and *describes* only

- *described by* some **Superman Comic** and
- *bearer of* some **Super Strength** and
- *has origin* value Krypton and
- *fires eye lasers* some **Laser** and …

Superman Example

<u>Superman</u> instance of **Fictional Description** and *describes* only

- *described by* some **Superman Comic** and
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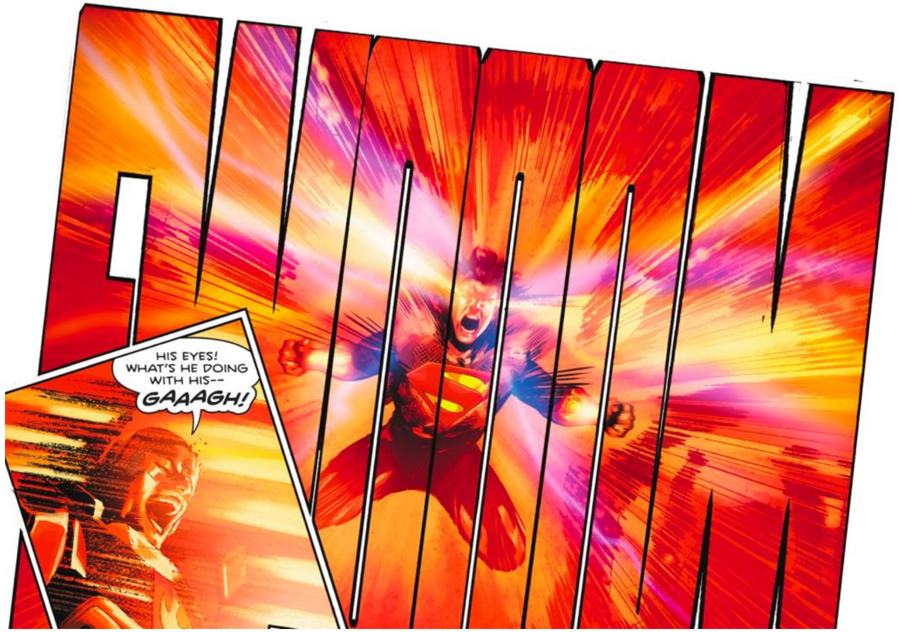
- *has origin* value <u>Krypton</u> and
- *fires eye lasers* some **Laser** and …

?!?!?!

?!?!?!



Superman Example

<u>Krypton</u> instance of **Fictional Description** and *describes* only

- **Astronomical Entity** and
- *bearer of* some **Rocky Quality** and …

x *fires eye laser* y iff x instance of **Person** and *has part* some (**Eye** and *bearer of* **Laser Firing Disposition**) and y instance of **Laser** and …

Cyber War Game Simulation Example

Red team simulation representation instance of **Cyber War Game Simulation Representation** and *represents* only

- **Cyber-Attack Process** and
- *has participant* value US Army and
- *has participant* **Adversary** and
- *has occurrent part* (**Act of Targeting** and has value Army Network 1 and *occupies temporal region* value March 23 2025) and
- *has occurrent part* **Strategic Response Process** and (*has participant* value US Army Cyber Response Team) and (*occupies temporal region* value March 24, 2025) and …

Recipe

- Introduce an **information content entity** (ICE); identify classes and object properties reflecting intended meaning of the ICE such as **super strength**, *continuant part of*, and so on

- Assert ICE *is about* **only** a class C equivalent to the logical combination of classes and restrictions on object properties where:
  - Fictions *describe*
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  - Simulations *represent*

- For class or relata of any object property constituting C with no instance, return to second step; otherwise, stop

Upshot

- Proper understanding of fiction, simulations, and blueprints favors interpretations in which relevant information is about logical combinations of types
- Combining this with realism, we maintain the target of relevant information is ultimately real types
- We provide a recipe that can be represented in OWL to thereby deploy

Looking Ahead

- This lecture was placed first for a reason; I want you to reflect on the nuance as you listen to subsequent discussions today regarding representing the law and systems

- We will revisit this topic during the exercises where you will be given approximately to hours to build design patterns and encode representations of unreal entities